SOP-118 — Equipment Isolation and Lock-Out Tag-Out

Plant Site A · Rev 6 · Issued 30/04/2025 · Approved by Plant Engineering

1. Purpose

This procedure defines the mandatory method for isolating plant equipment from all energy sources before intrusive work, and for locking and tagging those isolations so they cannot be defeated while people are exposed. Isolation discipline is the single most important barrier against fatal accidents during maintenance at Plant Site A.

2. Scope

Applies to every intrusive maintenance, inspection or cleaning task on registered equipment, including pumps, heat exchangers, control valves and their associated piping. It covers electrical, process, hydraulic and stored mechanical energy. Work on live electrical equipment is prohibited and is not enabled by this procedure.

3. References

Site isolation standard and the plant single line diagrams.

Piping and instrument diagrams for the system containing the equipment.

Permit to work procedure for Plant Site A.

4. Safety Precautions

Only authorised isolators listed on the site register may apply or remove isolations. Training does not confer authorisation; the register does.

Every person working under an isolation shall fit their personal lock to the lockbox. No lock, no work, without exception.

An isolation proven dead at the start of work is proven for that moment only. If the job is suspended overnight or the scope changes, the isolation shall be re-proven before work resumes.

Never assume a valve position from its handwheel or lever. Verify by system response, vent behaviour or physical disconnection, because valve position indicators fail and gearbox drives strip without external sign.

5. Tools and Materials

Have available: the site padlock series with unique keys, the lockbox, isolation tags and certificate forms, chains and valve covers for locked closed valves, rated blanking spades and bleed rings where positive isolation is specified, a voltage tester proved against a known live source for electrical proving, and the piping and instrument diagrams marked with the isolation points.

Personal locks are individually issued and shall never be shared, transferred or removed by anyone other than the owner, except under the site lost-lock procedure administered by the safety department.

6. Procedure

6.1 Plan the isolation on the piping and instrument diagram before going to the field. List every energy source: driver electrical supply, process connections, flushing connections, instrument air to actuated valves, and any stored energy such as sprung actuators or elevated liquid legs.

6.2 Prepare the isolation certificate listing each isolation point with its unique valve or breaker identifier, the required state, and the securing method, whether lock, chain or blank.

6.3 Isolate the driver electrically by racking out the breaker or withdrawing the fuses, and lock the switching device in the isolated position. Pressing the local start button is part of proving, never a substitute for electrical isolation.

6.4 Isolate process connections by closing the identified valves and locking them closed. For entry into equipment or hot work, positive isolation by spade or physical disconnection is required; a closed valve alone is not positive isolation.

6.5 Treat instrument air to spring return actuators, elevated liquid legs, accumulators and trapped thermal expansion as energy sources in their own right: isolate, vent or drain them on the certificate, and physically restrain any actuator that could move when air is removed.

6.6 Drain and vent the equipment to the safe location identified on the certificate, and leave drains and vents open and tagged for the duration of the work.

6.7 Prove dead at the point of work: attempt a local start for electrical isolation, and verify zero pressure at the equipment vent for process isolation, in the presence of the person doing the work.

6.8 Fit the isolation tags at every point, place the isolation keys in the lockbox, and have every member of the working party fit a personal lock before first tool contact.

6.9 Where two or more work parties share an isolation, each party fits its own lock to the common lockbox, and the isolation shall not be broken until every party has signed off and removed its lock. The isolator owns the count.

6.10 Long term isolations spanning more than one shift shall be revalidated at each shift handover by the oncoming authorised isolator, walking the points against the certificate and initialling the validation record.

6.11 On completion, the isolator shall walk the isolation list in reverse, confirming the work site is clear, guards are refitted and every person has removed their lock, before restoring any energy source.

6.12 Return the equipment to the operating team formally, and file the completed isolation certificate against the work order for the equipment tag.

7. Acceptance Criteria

An isolation is acceptable only when every energy source on the certificate is isolated, secured and proven dead at the point of work, and every worker is protected by a personal lock. Restoration is complete only when the certificate is signed off and the operating team has accepted the equipment back.

8. Records

Completed isolation certificates are retained with the associated work order for two years. Any isolation defeated, missing or found in the wrong state shall be reported as a high potential incident and investigated. The site isolation register shall be audited quarterly by the safety department, sampling live isolations in the field against their certificates.